

Radar-Informed 3D Multi-Object Tracking under Adverse Conditions

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Abstract—The challenge of 3D multi-object tracking (3D MOT) is achieving robustness in real-world applications, for example under adverse conditions and maintaining consistency as distance increases. To overcome these challenges, sensor fusion approaches that combine LiDAR, cameras, and radar have emerged. However, existing multi-modal fusion methods usually treat radar as another learned feature inside the network. When the overall model degrades in difficult environmental conditions, the robustness advantages that radar could provide are also reduced. In this paper we propose RadarMOT, a radar-informed 3D MOT framework that explicitly uses radar point cloud data as additional observation to refine state estimation and recover detector misses at long ranges. Evaluations on the MAN-TruckScenes dataset (TruckScenes) show that RadarMOT consistently improves the Average Multi-Object Tracking Accuracy (AMOTA) by 12.7% at long range and up to 10.3% in adverse weather. The code will be available at <https://github.com/bingxue-xu/radarmot>

Index Terms—3D Multi-Object Tracking, Radar Association, Autonomous Driving

I. INTRODUCTION

Safe 3D perception requires a system to continuously understand *what* is around it and *where they are going* before acting. This is the task of 3D MOT: maintaining identities and motion trajectories of detected objects across time. Trajectory prediction and motion planning depend directly on the quality of these estimates. 3D MOT operates by detecting objects independently in each frame and then associating them across time to form trajectories. This makes 3D MOT performance fundamentally dependent on two things: (i) *how accurately their motion is estimated* (state estimation accuracy), and (ii) *how reliably detections are matched with tracked objects* (data association robustness).

Radar has regained attention because it is weather-resistant and works reliably in poor visibility conditions, making it a good complement to LiDAR and cameras exactly where those sensors are weakest. Recent radar fusion methods [1]–[4] tend toward data-driven, mostly early-fusion strategies that learn radar features within the detection network. This is mainly due to: 1) detection primarily determining overall tracking performance, and 2) radar point clouds are sparse and noisy. Multimodal fusion methods [2], [5]–[7] explicitly use radar-refined detection velocity in data association to improve tracking performance. Despite achieving strong results on tracking benchmarks, these feature-fusion approaches still depend primarily on geometric cues from LiDAR or semantic cues

from images. When these two sensors struggle under adverse weather, the fused representations also become unreliable, and the models undermine radar’s inherent advantages.

To reduce the computation burden on the graphics processing unit (GPU), we revisit both the radar sensing mechanism and Kalman filter-based 3D MOT frameworks [8]–[12]. Ther, we first incorporate radar point clouds as an additional measurement in the Kalman filter update, which consistently boosts AMOTA and lowers false positives, indicating a more accurate state estimation. Next, we address motion blur during data preprocessing by leveraging radial velocity. Based on the observation that the distance-based tracker CenterPoint [13] yields more true positives than MCTrack [9], we further extend data association into a two-stage scheme that utilizes bidirectional distances with radar association, thereby stabilizing the tracking system.

RadarMOT achieves strong and consistent improvements on TruckScenes, demonstrating that leveraging radar physical measurements in the tracking stage can substantially increase robustness when LiDAR and cameras degrade. Overall, RadarMOT improves glsamota by +6.7% over the MCTrack baseline and reduces glsids by 30%, with the largest gains appearing at long range and under adverse conditions. The contributions of this paper are:

- We propose **RadarMOT**, a radar-informed 3D MOT framework that integrates radar as an explicit observation for tracking without deep learning, demonstrates robustness under adverse conditions and at long ranges.
- We introduce a practical **motion compensation** pipeline for multi-sweep radar aggregation that accounts for both ego motion and target motion, compensates temporal offsets using radar Doppler measurements.
- We formulate a **radar-informed Kalman filter update** that uses associated radar radial velocities to refine tracked objects’ planar velocity and stabilize trajectories.
- We propose a **two-stage association** strategy with cross-check association followed by radar association to reduce mismatches and recover detector misses, particularly at long range.

II. RELATED WORK

3D MOT is dominated by the geometry-based Tracking-by-Detection (TBD) paradigm and the learning-based Tracking-by-Attention (TBA) paradigm. Most radar-based multimodal fusion approaches fall within the TBA category.

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Tracking by Detection TBD divides the tracking task into four independently modules: detection, motion model, data association, and lifecycle management [11]. Since AB3DMOT [8] established a strong baseline by coupling Kalman filter [14] with Intersection over Union (IoU)-based Hungarian matching [15]. The following methods have progressively refined the association strategy, from generalized IoU to normalization [16], lifting it to 3D [11], and combining distance with IoU [9]. Another representative approach, CenterPoint [13], replaces IoU with Euclidean distance combined with greedy nearest-point matching, achieving strong performance on nuScenes [17] and Waymo [18]. The following work [10] replaces Euclidean distance by Mahalanobis distance. Other methods [12], [19]–[21] enhanced the motion model within the Kalman filter. SimpleTrack [11] identifies non-maximum suppression (NMS) and proposes two-stage association. MCTrack [9], consolidates these advances into a unified and general framework that refines the Kalman filter and incorporates both IoU and distance into the data association stage, achieving state-of-the-art performance across the KITTI [22], nuScenes [17], and Waymo [18] datasets. We therefore adopt MCTrack as our baseline.

Tracking by Attention Emerging TBA approaches replace explicit associations with learned queries that carry object identity across frames. Recent 3D MOT methods [5], [23]–[28] encode tracked objects as queries, which attend to features of the current frame to both update object states and detect new objects in a single step, eliminating the need for hand-crafted cost metrics or lifecycle heuristics. Early approaches [23], [29] introduced the joint detection-and-tracking framework by propagating object queries over time and preserving identities via attention. Following works [30], [31] extended this idea to longer temporal modelling through track-aware supervision and improved training schemes. Subsequent works [30]–[32] skip the intermediate output detection and introduce end-to-end tracking frameworks. Despite these advances, a common weakness of TBA methods is tight coupling. If the detector or features degrade (bad weather, night, long range, sparse points), tracking can degrade too, and changing sensors usually means retraining.

Radar-based Tracking Radar has recently regained attention due to its robustness to adverse weather, particularly in the scope of camera-radar fusion [33], [34]. A dominant direction incorporates radar at the detection stage before tracking [1]–[4], [35] and improves 3D MOT performance by radar-enhanced detection. References [5], [6], [36], [37] train models using radar velocity as input and employ the predicted bounding box velocity in motion-aware association. DopplerTrack [38] revisits Doppler velocity for moving object tracking, highlighting the potential of radar for motion-centric tracking. RaTrack [36] presents a radar-only method for tracking moving objects that avoids relying on 3D bounding boxes and object classification by using a multi-task network. Similarly, our approach performs motion segmentation, but we extract moving objects via geometric filtering, thus bypassing the need for deep learning.

III. METHOD

We propose RadarMOT, as shown in Fig. 1. It builds upon MCTrack [9] and incorporates radar data. We treat radar data as additional observations alongside 3D detection bounding boxes. The radial velocity is used to refine state estimation in the Kalman filter, and the radar points indicate object presence when objects are temporarily missed by the detector. This direct design increases the robustness of tracking when detection is weak and compatible with any detector.

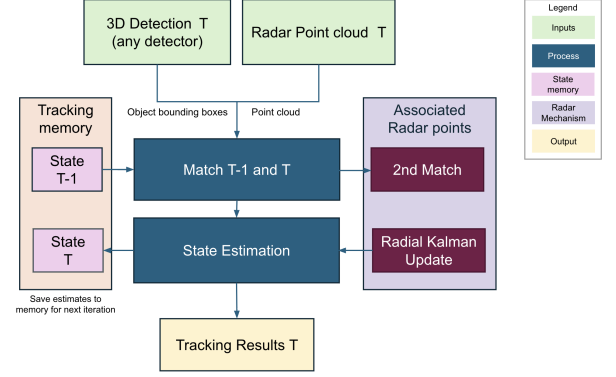


Fig. 1: Overall framework of the proposed RadarMOT, built upon MCTrack [9] and incorporating radar. It first associates tracked objects from both detection and radar in two stages, and then refines the state estimate by a radial velocity-constrained Kalman filter.

A. Motion Compensation

We observed the motion blur problem caused by the ego motion and target motion while aggregating multi-sweep radar data, and addressed this displacement using both ego motion and radar radial velocity.

Ego motion compensation Following the ego-motion compensation [39], we observe that static objects still have residual radial velocities when the vehicle rotate. Therefore we also account for the ego rotation. By doing so, we correctly remove both the translational and rotational ego-motion effects. Let i denotes the i th radar point, $\mathbf{b}_i$ denotes the line-of-sight unit vector of radar point p_i . After ego motion compensation, the point velocity in the reference coordinate system is written as:

$$\mathbf{v}_{\text{comp}}^{(i)} = \mathbf{v}_{\text{rel}}^{(i)} + \mathbf{v}_{\text{ego}} + \boldsymbol{\omega}_{\text{ego}} \times \mathbf{r}^{(i)} \quad (1)$$

where $\mathbf{v}_{\text{rel}}^{(i)}$ denotes the radar-reported relative velocity of point i , $\mathbf{v}_{\text{ego}}$ is the ego linear velocity, $\boldsymbol{\omega}_{\text{ego}}$ is the ego angular velocity, and $\mathbf{r}^{(i)}$ is the vector from the ego rotation center to the radar point.

Radar motion compensation After ego-motion compensation, dynamic objects can still appear displaced because radar detections are acquired at the sweep timestamp rather than at the keyframe timestamp. Unlike LiDAR methods that infer object motion via scene flow [40], radar Doppler directly measures velocity, capturing non-ego motion without learning. To mitigate the temporal offset, each radar point is translated

according to its estimated motion over the time interval between the sweep timestamp t_s and the keyframe timestamp t_k . The resulting motion-compensated point position is given by:

$$\mathbf{p}_{\text{comp}}^{(i)} \approx \mathbf{p}_k^{(i)} + \Delta r^{(i)} \mathbf{b}_i^{(i)} \quad (2)$$

where $\mathbf{p}_k^{(i)}$ denotes the point position after transformation to the keyframe in the reference coordinate system, $\mathbf{b}_i^{(i)}$ is the radar line-of-sight unit vector, and $\Delta r^{(i)} = v_{\text{radial}}^{(i)}(t_k - t_s)$ is the estimated radial displacement during the time offset.

Fig. 2 shows the effect of the proposed motion compensation in the bird's-eye view (BEV). Because radar measures only radial velocity, tangential velocity is unobservable, causing residual distortion, especially for objects crossing the radar's line of sight. This effect is strongest near the ego vehicle, so a nearby region is disabled (15 m ego-centric radius in this implementation).

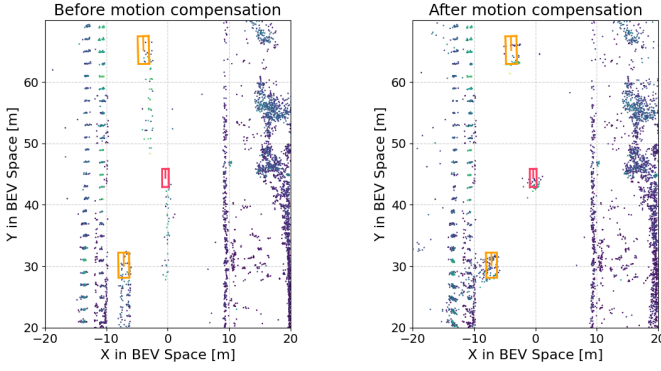


Fig. 2: Comparison of radar point clouds before and after motion compensation. Bounding boxes indicate annotated ground truth.

B. Radar Informed Kalman filter

To update state estimation with radar information, we treat the associated radar radial velocity as an additional observation in a Kalman filter, thereby refining the tracked objects' planar velocity and indirectly stabilizing their positions over time.

In short, for each tracked object we (i) associate radar points that fall inside an inflated track box and have consistent line-of-sight velocity with the box motion (4), (ii) build an observation matrix $\mathbf{H}_{r,k}$ whose rows project the track's planar velocity onto each radar point's line-of-sight direction (5)(7), and (iii) update the state with the innovation between measured and predicted radial velocities (9). This uses radar primarily to constrain velocity, which improves tracking robustness under occlusion and reduces velocity drift.

Motion Model We use a constant-velocity motion model. Let k denotes the keyframe index. The state $\mathbf{x}_k = [x_k, y_k, v_{x,k}, v_{y,k}]^\top$ represents the object position and planar velocity in the global coordinate system. $\mathbf{F}$ denotes the state transition matrix, $\mathbf{Q}_k$ denotes the process noise. The predicted state is then given as:

$$\mathbf{x}_{k|k-1} = \mathbf{F}\mathbf{x}_{k-1|k-1} + \mathbf{w}_k \quad \mathbf{w}_k \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}_k) \quad (3)$$

Radar Observation Model To handle noisy radar points, only radar points reflecting from the same object are used as observations. Let p_i denote the i -th radar point, and let $\mathbf{p}_s^{(i)} \in \mathbb{R}^3$ denote its position, with measured radial velocity $v_{\text{radar}}^{(i)} \in \mathbb{R}$, and let $v_{\text{box}}^{(i)} \in \mathbb{R}$ denote the bounding box velocity projected onto the line-of-sight of point i . The associated radar point set is defined by two filters:

$$\mathcal{R}_{\text{assoc}} = \{ p_i \in \mathcal{R}_k \mid \mathbf{p}_g^{(i)} \in \alpha \mathcal{B}_k^t \wedge |v_{\text{radar}}^{(i)} - v_{\text{box}}^{(i)}| \leq \delta_v \} \quad (4)$$

where

- $\alpha \mathcal{B}_k$ restricts radar points to the track bounding box in XY plane, α denotes the inflation rate.
- δ_v removes outliers based on the radial velocity difference between each radar point $v_{\text{radar}}^{(i)}$ and the bounding box velocity $v_{\text{box}}^{(i)}$ projected onto the radar point's line-of-sight direction.

To construct the radar observation matrix $\mathbf{H}_{r,k}$, where r means observation from radar at the k -th frame, the line-of-sight unit vector of each associated radar point i is defined from its position $\mathbf{p}_s^{(i)} \in \mathbb{R}^3$ in the radar sensor coordinate system S_i as:

$$\mathbf{b}_i = \frac{\mathbf{p}_s^{(i)}}{\|\mathbf{p}_s^{(i)}\|} \in \mathbb{R}^3 \quad (5)$$

The observation function h_i for point i projects the track velocity onto this line-of-sight direction, defined as:

$$h_i(\mathbf{x}_k) = \mathbf{h}_i^\top \mathbf{x}_k, \quad \mathbf{h}_i^\top = [0 \quad 0 \quad \mathbf{b}_i^\top S_i \mathbf{R}_{\text{global},xy}] \in \mathbb{R}^{1 \times 4} \quad (6)$$

where $\mathbf{b}_i \in \mathbb{R}^3$ is the line-of-sight direction in the radar coordinate system, $S_i \mathbf{R}_{\text{global},xy} \in \mathbb{R}^{2 \times 3}$ extracts the planar rows of the global-to-radar rotation, and the position columns of $\mathbf{h}_i^\top$ are zero since only velocity enters the projection. Stacking M such rows gives the joint observation matrix in (7).

$$\mathbf{H}_{r,k} = \begin{bmatrix} \mathbf{h}_1^\top \\ \vdots \\ \mathbf{h}_M^\top \end{bmatrix} \in \mathbb{R}^{M \times 4} \quad (7)$$

Let $\mathbf{z}_k^r = [z_1, z_2, \dots, z_M]^\top \in \mathbb{R}^M$ denote the vector of radial velocities measured by the M associated radar points. Following the standard linear observation model [41], the joint radar observation model is as:

$$\mathbf{z}_k^r = \mathbf{H}_{r,k} \mathbf{x}_k + \mathbf{n}_k, \quad \mathbf{n}_k \sim \mathcal{N}(\mathbf{0}, \mathbf{R}_{r,k}) \quad (8)$$

where $\mathbf{x}_k$ is the true object state, and $\mathbf{R}_{r,k} \in \mathbb{R}^{M \times M}$ denotes the measurement noise covariance.

Radar-Informed Kalman Filter Update Given the radar observation, the tracking state is then updated using Kalman filter, as illustrated in Fig. 3, the radar-informed state is updated by:

$$\begin{aligned} \boldsymbol{\nu}_k &= \mathbf{z}_k^r - \mathbf{H}_{r,k} \mathbf{x}_{k|k-1} \\ \mathbf{x}_{k|k} &= \mathbf{x}_{k|k-1} + \mathbf{K}_k \boldsymbol{\nu}_k \end{aligned} \quad (9)$$

where

- ν_k denote the innovation derived from the difference between the observed radial velocity and the motion model predicted radial velocity.
- z_k^r denotes the observed radial velocity.
- $H_{r,k}x_{k|k-1}$ denotes the track state estimate predicted radial velocity.
- $x_{k|k-1}$ and $x_{k|k}$ denote the predicted and updated track states.
- K_k is the Kalman gain.

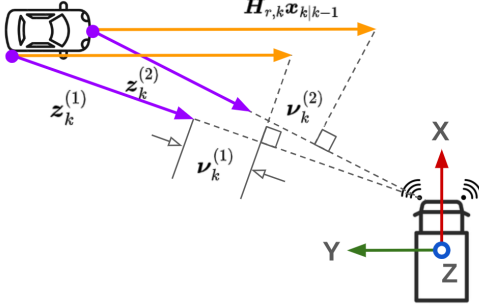


Fig. 3: Radar-informed Kalman update. Purple points and arrows represent radar points and motion-compensated velocity. Orange arrows represent the predicted velocity coming from the bounding box.

C. Two-Stage Association

To reduce mismatches, which are the primary cause of identity switches, we propose a two-stage association that first performs a bidirectional cross-check association and then applies a radar association.

Cross-Check Association Instead of associating at a single timestamp (e.g., either comparing the forward prediction with the detection at time t , or comparing the track with the backward predicted detection at time $t-1$), we use both forward and backward predictions for mutual cross-checking and combine a speed similarity measure to further minimize the influence of object orientation. The cost function is defined as:

$$\text{cost} = \alpha \frac{\|p_t^{\text{trk}} - p_t^{\text{det}}\|}{v_t^{\text{trk}} \Delta t} + \beta \frac{\|p_{t-1}^{\text{trk}} - p_{t-1}^{\text{det}}\|}{v_{t-1}^{\text{det}} \Delta t} + \delta \frac{|v^{\text{trk}} - v^{\text{det}}|}{\max(v^{\text{trk}}, v^{\text{det}})} \quad (10)$$

where

- p_t^{trk} and p_t^{det} are the track and detection centers at time t .
- p_{t-1}^{trk} and p_{t-1}^{det} are the track center and backward-predicted detection center at time $t-1$.
- v^{det} and v^{trk} are the detection and track speeds, reducing the influence of orientation error.
- α , β , and δ are weighting factors.

Radar Association To recover the detector missed objects, unmatched tracked objects are associated with radar measurements in a second stage.

We first remove the points that are close to detections to avoid duplicate associations, and then select candidate radar

points for unmatched tracked objects using (4). In addition, we filter out stationary clutter using a minimum velocity. If more than $N_{\min}$ points are collected, the track is considered radar-seen for the current frame, and the radar-informed Kalman filter update (9) is applied to refine the state estimation.

IV. EXPERIMENTS SETUP

Datasets Experiments are conducted on the largest public radar dataset TruckScenes [39] for autonomous driving, which includes 747 scenes of 20 seconds each, with keyframes at 2 Hz. The sensor suite includes 4 cameras, 6 LiDAR units, and 6 4D radar sensors with full 360-degree coverage. It covers diverse weather conditions (rain, fog, snow), lighting situations (darkness, glare, twilight), and driving areas (highway, urban, rural). This dataset also reflects the dynamic, long-range perception and occlusion challenges: 40% of annotated objects move faster than 20 m/s, 50% of all annotations are beyond 75 m, and 39% of the objects have a visibility of less than 41% [39]. Thus, evaluation is conducted using this dataset to investigate robustness.

Evaluation Metrics We follow the official nuScenes tracking challenge [42] and report AMOTA, along with the numbers of true positives (TP), false positives (FP), false negatives (FN), and identity switches (IDS). For all metrics, true positives are obtained by applying a 2 m threshold to the 2D center distance on the ground plane. As this is the first 3D MOT work on TruckScenes, the classes follow nuScenes tracking challenges [42] and ranges are adopted from TruckScenes detection benchmark [39], i.e., bigger vehicles (car, truck, bus, trailer) are evaluated up to 150 m and smaller objects (pedestrian, motorcycle, bicycle) up to 75 m.

Detection Input We use the same 3D detection results obtained by CenterPoint [13] on the TruckScenes validation dataset. The detection model is reproduced following the training recipe in TruckScenes [39]. It's worth mentioning that CenterPoint [13] achieves 58% mean Average Precision (mAP) on nuScenes but only 33% mAP on TruckScenes validation set, indicating that the environmental conditions in TruckScenes are more challenging for 3D perception.

Baseline We use MCTrack [9] as baseline, as described in Section II. We report the MCTrack tracking results on the TruckScenes validation set in the second row of Table I. Additionally, we report the CenterPoint tracker in the first row of Table I.

V. RESULTS

We report our results on the TruckScenes validation set in Table I. The proposed RadarMOT achieves 33.3% in terms of overall AMOTA, an absolute improvement of 6.7% over the MCTrack baseline, and a relative 30% reduction in Identity Switches (IDS).

To the best of our knowledge, this is the first 3D MOT investigation on the TruckScenes, offering a conservative baseline for the community.

Long Range To reveal how RadarMOT behaves as range increases, we evaluate the performance across three range bins

TABLE I: Performance comparison on MAN-TruckScenes validation set with the same detector. (Bic., Motor, Ped., Tra., Tru.) denote (Bicycle, Motorcycle, Pedestrian, Trailer, Truck). Bold indicates best performance per column.

Method	Detector	AMOTA % $\uparrow$								TP $\uparrow$	FP $\downarrow$	FN $\downarrow$	IDS $\downarrow$
		Overall	Bic.	Bus	Car	Motor	Ped.	Tra.	Tru.				
CenterPoint [13]	CenterPoint [13]	22.8	9.3	0.0	51.1	23.5	37.7	0.9	37.2	37149	8467	26912	7278
MCTrack [9]	CenterPoint [13]	26.6	11.9	0.0	40.0	59.4	36.4	0.7	37.6	33636	13026	32378	5325
RadarMOT (ours)	CenterPoint [13]	33.3	17.0	0.0	53.6	56.2	46.5	13.0	46.8	42717	12257	24906	3716

up to 150 m, as shown in Table II and Fig. 4. The growing margin reflects radar’s increasing contribution as LiDAR point density drops with range. Our method achieves an absolute +12.7% AMOTA improvement at the third bin (100–150 m) compared to baseline MCTrack.

TABLE II: Comparison across ranges on TruckScenes validation set. All methods use the same CenterPoint [13] detector. Bold indicates best value per column.

Method	Overall	I: 0–50 m	II: 50–100 m	III: 100–150 m
	AMOTA % / IDS	AMOTA % / IDS	AMOTA % / IDS	AMOTA % / IDS
CenterPoint [13]	22.8 / 7278	28.1 / 2766	21.5 / 3107	21.2 / 1518
MCTrack [9]	26.6 / 5325	30.8 / 1804	23.8 / 1704	20.1 / 836
RadarMOT (ours)	33.3 / 3716	36.0 / 1321	32.2 / 1451	32.8 / 895

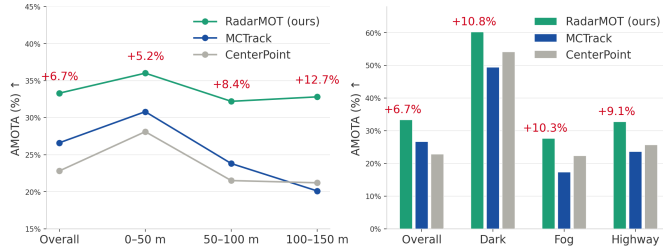


Fig. 4: Comparison across range and adverse conditions on TruckScenes validation set.

Adverse Conditions We also evaluate the performance under adverse conditions in different weather, lighting, and driving areas, see Table III and Fig. 4. RadarMOT achieves consistent absolute AMOTA improvements over the baseline MCTrack: +10.3% in fog, +10.8% at night, and +9.1% on highways. Contrary to our initial hypothesis, RadarMOT performs worse than the baseline in snow.

Ablation Study To isolate the contribution of each component, we add them incrementally on the baseline MCTrack [9], as shown in Table IV. Radar Kalman filter refinement achieves the lowest false positives and reduces IDS. Adding radar association captures more true positives but also increases false positives. This trade-off is stabilized by the cross-check association, which boosts additional true positives, highlighting that robust association is key before refinement.

VI. CONCLUSION

We present a direct radar fusion method for 3D MOT which does not depend on deep learning. It integrates radar data as additional observations and performs robustly under adverse

TABLE III: Comparison under adverse weather, lighting conditions and driving areas on TruckScenes validation set. Bold indicates the best value per row per metric.

Group	Condition	CenterPoint [13]	MCTrack [9]	RadarMOT (ours)
		AMOTA % / IDS	AMOTA % / IDS	AMOTA % / IDS
Weather	Clear	36.0 / 3040	41.4 / 2454	46.0 / 1552
	Overcast	24.2 / 2753	29.0 / 1775	33.6 / 1307
	Rain	20.9 / 1458	18.3 / 839	26.1 / 771
	Fog	22.3 / 173	17.3 / 109	27.6 / 76
	Snow	16.2 / 38	20.0 / 34	17.8 / 25
Lighting	Dark	54.1 / 1388	49.4 / 674	60.2 / 728
	Illuminated	22.7 / 4627	33.0 / 2952	38.8 / 2317
	Twilight	25.8 / 1095	26.8 / 906	31.0 / 508
	Glare	16.3 / 494	13.2 / 361	16.8 / 219
Area	Highway	25.7 / 5833	23.6 / 3975	32.7 / 2702
	City	21.5 / 887	24.4 / 687	27.4 / 577
	Rural	16.4 / 463	19.1 / 377	23.3 / 303
	Residential	25.3 / 53	29.6 / 83	28.6 / 54
	Terminal	51.9 / 33	56.5 / 9	56.6 / 13

TABLE IV: Ablation study on TruckScenes validation set. Each row adds one component on top of the MCTrack [9] baseline. Bold indicates best results.

Component			AMOTA % $\uparrow$	TP $\uparrow$	FP $\downarrow$	FN $\downarrow$	IDS $\downarrow$
Radar Kalman filter	Radar Association	Cross-Check Association					
			26.6	33636	13026	32378	5325
✓			29.2	33948	12126	32234	5157
✓	✓		30.7	38446	12861	26515	6378
✓	✓	✓	33.3	42717	12257	24906	3716

conditions and at long ranges. As the first 3D MOT study conducted on the TruckScenes dataset, we release our code to offer a solid baseline for the community. Thanks to the raw data fusion strategy, RadarMOT is practically suitable for robotics and autonomous driving applications.

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